Yani Aliche

Olathe, KS | (913)-286-9882 | yanialiche07@gmail.com | www.linkedin.com/in/yanialiche
Engineering Portfolio: **https://yanialiche.github.io/**

EDUCATION

Cornell University, Duffield College of Engineering, Ithaca, NY
B.S. in Mechanical Engineering, Minor Aerospace Engineering *Aug 2025 – May 2029*
Relevant Coursework: Engineering Operations: Data Science and Decision Making, Statics & Mechanics of Solids, Differential Equations for Engineers, Thermodynamics, Physics II: Electromagnetism; *Spring 2027*: Linear Algebra, Dynamics, Introduction to Mechanical Design

SKILLS

Technical: Fusion 360, SolidWorks, AutoCAD, Microsoft Office Suite (Excel, Word, PowerPoint), Tolerance Analysis
Programming/Analysis: Python, MATLAB, ANSYS (Mechanical/Fluent)
Language: English (Native), French (Intermediate), Tamazight (Native)

EXPERIENCE

Cornell Design Build Fly | Aerodynamics and Controls Sub-team *Sept 2026 – Present*

- Designed and modeled the aircraft empennage and tail control surfaces in SolidWorks, using component analysis and testing to refine designs for manufacturing
- Used ANSYS Fluent for CFD simulations and MATLAB to analyze results, evaluate tail aerodynamic performance, and guide design decisions
- Collaborate on the design, manufacturing, and testing of an aircraft for the AIAA Design Build Fly competition

PROJECTS & RESEARCH

Independent Research: Airfoil Surface Design and Aerodynamic Performance *Sept 2024 – May 2025*

- Fabricated and adapted 5 3D-printed airfoils with varied surface patterns to evaluate aerodynamic efficiency
- Analyzed lift and drag data using experimental design methods to identify performance differences
- Presented at the Greater KC Science & Engineering Fair; awarded *1st Place, Engineering Mechanics Division*

Airfoil Design & Analysis Project | SolidWorks *Aug 2025 – Oct 2025*

- Modeled research airfoils in SolidWorks, extending prior experimental work into a computational framework
- CFD simulations to evaluate lift and drag and structural analysis to assess wing stress under aerodynamic loads

Open-Fan Engine Design Project | Fusion 360 & SolidWorks *Feb 2026 – Mar 2026*

- Designed a conceptual open-fan jet engine (inspiration from CFM RISE program) using Fusion 360/ SolidWorks
- Made design decisions about component geometry and placement to get a complete open fan engine concept

Automotive Powertrain Design | SolidWorks *July 2026*

- Designed and assembled an automotive powertrain with a engine, driveshaft, and differential assemblies
- Built assembly modeling skills by integrating individual parts into a complete automotive powertrain using mechanical mates in SolidWorks

ACTIVITIES & LEADERSHIP

Intramural Sports Supervisor – Cornell Recreational Services *May 2026 – Present*

- Lead intramural soccer, basketball, and volleyball operations for 200+ weekly participants
- Mentor and train new referees on rules, game control, and conflict resolution
- Support staff leadership in resolving disputes, evaluating referees, and maintaining safety standards

American Society of Mechanical Engineers (ASME) | Student Member *Oct 2026 – Present*

CREDENTIALS

- **Certified SOLIDWORKS Associate (CSWA)** *Sept 2026*
- Cornell Tradition Fellow *Aug 2025 – Present*
- OIE Engineering Scholar *Aug 2025 – Present*
- GKCSEF Engineering Mechanics — Senior Division Winner *Mar 2025*